Theodoros Tsilivis

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EDUCATION

Boston University **Boston, MA | 2023–Present | Expected May 2028**
Ph.D. Candidate in Computing and Data Science
Advisor: Kira Goldner

National Technical University of Athens **Athens, Greece | 2016–2022**
B.Sc. and M.Sc. in Electrical and Computer Engineering
GPA: 8.95/10 (top 5%). Thesis: "Learning Augmented Combinatorial Auctions."
Advisors: Dimitris Fotakis (NTUA) and Vasilis Gkatzelis (Drexel University).

RESEARCH INTERESTS

Algorithmic mechanism design in modern learning environments, including interdependent values, AI-driven markets, and learning-augmented mechanisms.

PUBLISHED AND ONGOING WORK

All author lists are in alphabetical order.

- Kira Goldner, Taylor Lundy, and **Thodoris Tsilivis**. "A Constant-Factor Approximation to Multidimensional Consumer Utility." *Preprint on arXiv.*
- Alon Eden, Kira Goldner, Eldar Kerner, and **Thodoris Tsilivis**. "From Welfare to Utility: Generalized Objectives in Budget-Feasible Procurement." *ICML 2026.*
- Kira Goldner, Divyarthi Mohan, and **Thodoris Tsilivis**. "Knowing Who, Not How Much: Learning-Augmented Mechanisms for Consumer Utility Maximization." *ICML 2026.*
- Shahar Dobzinski, Alon Eden, Kira Goldner, Ariel Shaulker, and **Thodoris Tsilivis**. "Bilateral Trade with Interdependent Values: Information vs. Approximation." *EC 2025.*

AWARDS

- Wexler Fellowship (2023-24).
- Gerondelis Foundation Fellowship (2024-25).
- Eleni Gatzoyiannis Scholarship (2024-25, 2025-26).

INVITED TALKS, POSTERS, AND PARTICIPATION

- **Talk:** "Generalized Objectives in Budget-Feasible Procurement," JECCO UK, Oxford, July 2026.
- **Talk:** "Bilateral Trade with Interdependent Values," Northeastern Theory Seminar, Boston, February 2026.
- **Talk:** "Bilateral Trade with Interdependent Values," Harvard EconCS Seminar, Boston, February 2026.
- **Poster:** "Generalized Objectives in Budget-Feasible Procurement," ICML, Seoul, July 2026.
- **Poster:** "Learning-Augmented Consumer Utility Maximization," ICML, Seoul, July 2026.
- **Poster:** "Bilateral Trade with Interdependent Values," Workshop on Algorithms, Learning, and Games (ALGA), Sardinia, June 2025.
- **Volunteer:** Boston Economics and Computing Hub (BEACH) workshops, Boston, Spring and Fall 2025.
- **Volunteer:** Workshop on Algorithms for Learning and Economics (WALE), Kefalonia, June 2024.

ACADEMIC EXPERIENCE

Teaching Assistant, Algorithms for Data Science, Boston University **Spring 2026**
Led discussion sections for 60 students; developed discussion, homework, and exam material.

Undergraduate Researcher and Teaching Assistant, NTUA **2021-2023**
At Corelab, co-organized the algorithms study group; assisted courses in Algorithms and Complexity, Discrete Mathematics, and Computer Programming.

REVIEWING

External reviewer for Easy Peasy at EC 2026 and Reliable ML from Unreliable Data at NeurIPS 2025; external subreviewer for SODA 2026.

TECHNICAL SKILLS

Theory: algorithmic game theory, optimization, randomized algorithms, approximation algorithms
Programming and tools: Python, C++, C, Bash, SQL, Git, Linux, Jupyter
Machine learning and data: NumPy, pandas, scikit-learn, PyTorch, TensorFlow/Keras, Matplotlib